



Your Avatar: The Way Of ML Part - I



- Jake models skimwing-taming success (1/0). Adding more features increases training accuracy but not validation accuracy. ☐
A. Model underfits B. Model overfits C. Threshold too low D. One-hot encoding needed **B**
- Dr. Garvin is modeling the length of a Tulkun (y) based on its age (x). Payakan is a massive outlier, he is young but huge. If Garvin uses Mean Squared Error (MSE) as the loss function compared to Mean Absolute Error (MAE), how does the regression line react? ☐
A. The line ignores Payakan entirely B. The line shifts significantly towards Payakan to minimize the squared residual C. The line stays exactly in the middle of the main cluster D. The line shifts away from Payakan **B**
- Training many trees on bootstrapped subsets to defend against Tulkun hunters mainly helps by: ☐
A. Bias reduction B. Variance reduction C. Both bias & variance reduction D. Increasing training loss **B**
- Lo'ak estimates Tulkun fin height parameters assuming Gaussian distributions. He should ideally: ☐
A. Maximize posterior probability B. Maximize empirical risk C. Maximize likelihood of observed data D. Maximize prior probability **C**
- A model warns if a Tulkun is in danger. False negatives are extremely dangerous; false positives are still tolerable. One should optimize: ☐
A. Accuracy B. Precision C. Recall D. Specificity **C**
- The Metkayina clan builds a classifier that predicts whether an approaching sea creature is friendly or hostile based on features like fin patterns, body glow, swimming trajectory, and ultrasonic signal. They use a very complex hypothesis class whose VC dimension is extremely large compared to the number of samples. What is the MOST likely consequence of deploying such a model ☐
A. Model has high bias B. Model has high variance / overfitting risk C. Model is always PAC-learnable D. Learning reduces ϵ magically **B**
- Colonel Quaritch uses a Decision Tree to find Jake. He checks two features: 'Is it a forest?' (Split A) vs. 'Is it an island?' (Split B). Split A results in two child nodes with high entropy (mixed targets). Split B results in one node being 100% pure (only Na'vi). Which split does the algorithm choose? ☐
A. Split A, because it keeps the tree balanced. B. Split B, because it yields higher Information Gain C. Split A, because it has higher Gini Impurity D. Neither, it chooses randomly **B**
- To decide on war, Tonowari asks 100 warriors to vote. Each warrior looks at a random subset of the evidence. The final decision is the majority vote. This reduces the variance of the decision. This is an example of: ☐
A. Boosting B. K-Nearest Neighbors C. Bagging D. Hierarchical Clustering **C**
- Tulkun pods are dense but have irregular, winding shapes (like snakes). They are not circular blobs. Which clustering algorithm works best to identify these pods? ☐
A. K-Means B. GMM C. DBSCAN D. Linear Regression **C**
- If Eywa knows the true distributions of all life perfectly, the error she achieves for her model is called the Bayes Error. Can we improve upon this error? ☐
A. Yes, by adding more data B. Yes, by more relevant data C. No, this is the irreducible error D. Yes, by using Bagging **C**

11. The Metkayina clan uses K-Nearest Neighbors to predict whether a newly spotted aquatic creature is dangerous by comparing its movement and fin-pattern features to known species with different distance metrics. They observe that two feature dimensions have very different numeric scales (e.g., fin length ≈ 1 -10, ultrasonic frequency $\approx 10,000$ -60,000). Which distance metric is MOST appropriate without explicit feature scaling?
- A. Euclidean distance B. Manhattan distance C. Cosine similarity D. Hamming distance **C**
12. Kiri wants to visualize 8,000-dimensional bioluminescent genetic signatures of ocean life. She first applies PCA $\rightarrow$ 80-dim representation, then tests t-SNE $\rightarrow$ 2-dim, observing well-separated clusters. However, when she directly applies t-SNE to the 8,000-dim raw data, the clusters overlap heavily. Why did the 2-step pipeline work better?
- A. PCA removed noise and highlighted global linear structure before t-SNE captured nonlinear local structure B. PCA ensures class separation while t-SNE only visualizes it C. t-SNE is not robust on high-dimensional data at all D. PCA forces t-SNE to use Euclidean distance instead of KL divergence **A**
13. The clan applies Apriori association analysis on inter-clan trade logs and discovers: “If Clan A trades whale bone jewelry AND medicinal kelp, $\rightarrow$ they almost always trade heat-resistant seashell armor”; Tonowari proposes using these rules directly in a recommender system that predicts the next resources desired by individual clans. Why might this fail in practice?
- A. Association rules cannot scale to large datasets B. Association rules assume independence, while recommender systems assume correlation C. Recommender systems require user-specific utility/affinity modeling, not population-level co-occurrences D. Recommender systems and association rules are mutually exclusive learning paradigms **C**
14. To predict warship movement patterns, Tonowari trains a logistic regression model using gradient descent (GD) on a large dataset of n samples and d features. He notices that increasing d significantly increases training time, even though n stays constant. Which statement best describes the per-iteration cost of GD responsible for the slowdown?
- A. $O(n+d)$ - gradient updates depend on both size and dimensionality equally B. $O(nd)$ - gradients require computing a full pass over the dataset with all features C. $O(n^2)$ - higher dimension forces quadratic gradient computation D. $O(d^2)$ - gradient update requires multiplying all feature pairs **B**
15. To detect enemy RDA submarines, Neytiri trains a Random Forest with T trees, each of maximum depth h , using n samples and d features. She notices that increasing the number of trees improves accuracy but dramatically increases training time. What is the closest estimate of the training complexity?
- A. $O(Tn \log n)$ - since each tree uses only a subset of data B. $O(Td^2)$ - feature sampling causes quadratic complexity C. $O(Tndh)$ - each tree is trained on roughly n samples with split tests across d features and depth h D. $O(n^2d)$ - because later trees correct mistakes from earlier trees **C**